English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Software Engineering, 13.12.2025 Test 1

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

Problem 1.

1. (0.25 points) A box contains 12 distinguishable bolts, five defective. Four bolts are chosen at random. What is the probability that no two defective bolts are adjacent in the order chosen?

2. (0.25 points) X, Y have the following joint distribution:

$Y \backslash X$	1	2	4
-1	$3/15$	$1/15$	0
0	α	2α	$3/15$
1	$1/15$	0	$1/15$

   Find α, the marginal distributions of X, Y, $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $Cov(X, Y)$ and $\rho_{X,Y}$.

3. (0.25 points) A factory produces items that are defective independently with probability 0.02. Inspect items until three defective ones are found. Let X be the number inspected. Find $\mathbb{P}(X > 100)$. Estimate the probability that fewer than 500 inspections are needed.

4. (0.25 points) Let $X \sim \mathrm{Po}(\lambda_1)$ and $Y \sim \mathrm{Po}(\lambda_2)$ be independent. Define $T := X + Y$. Find the conditional distribution and expectation of X given T.

Problem 2. (1 point) A young estate agent calls the numbers in a telephone directory in order[1] until selling a luxury property in Kapinova Ravnina. Each call succeeds with probability α. Let N count people who decline. The agent also asks those who decline for their email addresses to add to a mailing list. Given $N = n$, each of these n people supplies an email address with probability β, independently. If X is the number of addresses collected, identify its distribution and find its expectation and variance.

Problem 3. (1 point) In a game, a fair die is rolled. Initially, there are ten rolls available. A six awards five bonus rolls, and a five awards three. Bonuses may be awarded at most twice: after two awards (of either size), later fives or sixes give no extra rolls. Find the expected total number of rolls.

Problem 4. A flamingo jumps along the integers, starting at 0. At each step it uniformly chooses one of the next three integers (from $m \in \mathbb{N}_0$, it moves to $m+1, m+2, m+3$ with equal probabilities). Let X_n be its position after step $n \in \mathbb{N}$.

1. (0.25 points) Find the distribution of X_2 and calculate $\mathbb{P}(X_3 = 6)$.

2. (0.25 points) Find the expectation and variance of X_n.

3. (0.25 points) Find $\rho_{X_n, X_{n+k}}$. What happens as $k \to \infty$?

Let τ_m count the steps until the flamingo reaches $m \in \mathbb{N}$, and put $\tau_m = \infty$ if it never does.

4. (0.25 points) Calculate $\mathbb{P}(\tau_4 = 2)$, $\mathbb{P}(\tau_4 = 3)$ and $\mathbb{P}(\tau_4 < \infty)$.

5. (0.25 points) Calculate $\mathbb{E}[\tau_5 \mid \tau_5 < \infty]$.

[1] Assume infinitely many numbers.